

D899

Statistics for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A realty office open since 1998 reports home prices in a county: most homes sold for \$200,000 to \$300,000, but a few estates sold for over \$2 million. The office has nine agents. Which statement is true?

- A. The mean and median are always equal in a large data set.
- B. The mean is greater than the median, and the median better describes a typical home.
- C. The mean is best because it uses every sale.
- D. The median is greater than the mean.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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2.

After a fire drill cut a test short, a teacher added 5 points to every score. Before the change, the mean was 70 and the standard deviation was 8. The class has 31 students. What are the new mean and standard deviation?

- A. Mean 75, standard deviation 40.
- B. Mean 75, standard deviation 13.
- C. Mean 70, standard deviation 8.
- D. Mean 75, standard deviation 8.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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3. [TRANSFER — reversed stem / different variable]

Every value in a data set is multiplied by 1.5. The original standard deviation was 6. What is the new standard deviation?

- A. 9
- B. 7.5
- C. 6
- D. 13.5

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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4.

Delivery times for 40 drivers at a regional courier have $Q1 = 20$ minutes, a median of 26 minutes, and $Q3 = 30$ minutes. The courier recently added electric vans. Which delivery time is an outlier?

- A. 35 minutes
- B. 42 minutes
- C. 6 minutes
- D. 48 minutes

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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5. [TRANSFER — reversed stem / different variable]

A box plot's upper outlier fence is 57, and $Q3$ is 42. What is the IQR?

- A. 22.5
- B. 38
- C. 10
- D. 15

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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6.

A district-wide survey of 500 teens finds that those who eat breakfast have higher average grades. The survey was run online over two weeks. What can a teacher correctly conclude?

- A. Breakfast and grades are associated, but other factors could explain the link.
- B. The causal claim is valid because 500 teens is a large sample.
- C. There is no relationship, because only experiments can show relationships.
- D. Eating breakfast causes higher grades.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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7.

During the first week of school, a class records each student's ZIP code, number of siblings, height, and favorite sport. The data are entered in a shared spreadsheet. Which variable is categorical even though it is recorded as a number?

- A. Height.
- B. Number of siblings.
- C. ZIP code.
- D. None; any variable recorded as a number is quantitative.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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8.

A class tests whether a 1972 quarter from the teacher's desk is fair. In 50 flips it lands heads 34 times. They simulate 1,000 sets of 50 fair flips; only 3 sets have 34 or more heads. What is the best conclusion?

- A. $p \approx 0.003$, strong evidence that the coin is not fair.
- B. $p \approx 0.003$, which proves the coin is biased.
- C. $p \approx 0.68$, so the coin is fair.
- D. $p \approx 0.997$, so the coin is fair.

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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9. [TRANSFER — reversed stem / different variable]

In 500 simulated trials that assume a claim is true, a result at least as extreme as the observed one happened 60 times. What is the approximate p-value?

- A. 8.33
- B. 0.88
- C. 0.12
- D. 60

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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10.

In a class of 30, a survey taken in October shows $P(\text{plays soccer}) = 0.4$, $P(\text{in band}) = 0.3$, and $P(\text{both}) = 0.1$. What is $P(\text{soccer or band})$?

- A. 0.12
- B. 0.6
- C. 0.7
- D. 0.8

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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11. [TRANSFER — reversed stem / different variable]

$P(A \text{ or } B) = 0.75$, $P(A) = 0.5$, and $P(B) = 0.45$. What is $P(A \text{ and } B)$?

- A. 0.30
- B. 0.20
- C. 0.225
- D. 0.95

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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12.

Using foam dice from the supply closet, a student says a sum of 7 and a sum of 12 are equally likely, since each is just one possible sum. The class has a double period that day. What is the teacher’s best response?

- A. State that $P(7) = 6/36$ and move on.
- B. Agree, because each face of a fair die is equally likely.
- C. Roll the dice 10 times to prove it.
- D. Have students list all 36 outcomes and count the ways to get each sum.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A realty office ~~open since 1998~~ reports home prices in a county: most homes sold for \$200,000 to \$300,000, but a few estates sold for over \$2 million. ~~The office has nine agents.~~ Which statement is true?

Target word: mean vs. median under skew **Decides:** A few very high prices make the data right-skewed.

A	Absolute language	Size does not remove skew.
B	Correct	The expensive estates pull the mean up; the median resists them.
C	Adjacent category	Using every value is exactly why extreme values distort it.
D	Inverted operation	That happens with left skew, not a high tail.

2. Answer D Concept 2

~~After a fire drill cut a test short,~~ a teacher added 5 points to every score. Before the change, the mean was 70 and the standard deviation was 8. ~~The class has 31 students.~~ What are the new mean and standard deviation?

Target word: effect of adding a constant **Decides:** A shift moves the center but not the spread.

A	Inverted operation	Multiplies the SD by 5 instead of recognizing a shift.
B	Confusion set	Adds the 5 to the spread as well as the center.
C	Dropped element	Ignores the added points entirely.
D	Correct	Every score moves up 5; distances between scores do not change.

3. Answer A Concept 2 · transfer

Every value in a data set is multiplied by 1.5. The original standard deviation was 6. What is the new standard deviation?

Target word: effect of scaling on SD **Decides:** SD scales by $|k|$.

A	Correct	$6 \times 1.5 = 9$.
B	Inverted operation	Adds 1.5 instead of multiplying.
C	Dropped element	Scaling, unlike shifting, does change the spread.
D	Confusion set	6×1.5^2 scales like variance, not SD.

4. Answer D Concept 3

Delivery times for 40 drivers at a regional courier have $Q1 = 20$ minutes, ~~a median of 26 minutes,~~ and $Q3 = 30$ minutes. ~~The courier recently added electric vans.~~ Which delivery time is an outlier?

Target word: apply the $1.5 \times \text{IQR}$ rule **Decides:** $\text{IQR} = 10$; fences are 5 and 45.

A	Adjacent category	Above $Q3$ is not the same as beyond the fence.
B	Dropped element	Outlier only under a $1 \times \text{IQR}$ fence of 40; the 1.5 factor is dropped.
C	Confusion set	Lower fence is $20 - 15 = 5$; 6 is inside.
D	Correct	Upper fence = $30 + 15 = 45$; 48 is beyond it.

5. Answer C Concept 3 · transfer

A box plot's upper outlier fence is 57, and Q3 is 42. What is the IQR?

Target word: solve for IQR **Decides:** $57 - 42 = 1.5 \times \text{IQR}$.

A	Inverted operation	Multiplies by 1.5 instead of dividing.
B	Confusion set	$57 \div 1.5$ treats the fence as if it were $1.5 \times \text{IQR}$, ignoring Q3.
C	Correct	$15 = 1.5 \times \text{IQR}$, so $\text{IQR} = 10$.
D	Dropped element	That is $1.5 \times \text{IQR}$; it still needs dividing by 1.5.

6. Answer A Concept 4

A district-wide survey of 500 teens finds that those who eat breakfast have higher average grades. ~~The survey was run online over two weeks.~~ What can a teacher correctly conclude?

Target word: scope of conclusion **Decides:** No treatment was assigned; this is observational.

A	Correct	Without random assignment, confounders such as sleep or family routine remain possible.
B	Camouflage	Sample size does not fix confounding.
C	Absolute language	Observational data can show association, just not cause.
D	Confusion set	Treats an observed association as cause and effect.

7. Answer C Concept 5

~~During the first week of school,~~ a class records each student's ZIP code, number of siblings, height, and favorite sport. ~~The data are entered in a shared spreadsheet.~~ Which variable is categorical even though it is recorded as a number?

Target word: variable type **Decides:** Averaging ZIP codes is meaningless.

A	Adjacent category	A measurement is quantitative (continuous).
B	Confusion set	A count is quantitative (discrete).
C	Correct	ZIP codes are labels; arithmetic on them makes no sense.
D	Absolute language	Numeric labels are still categories.

8. Answer A Concept 6

~~A class tests whether a 1972 quarter from the teacher's desk~~ is fair. In 50 flips it lands heads 34 times. They simulate 1,000 sets of 50 fair flips; only 3 sets have 34 or more heads. What is the best conclusion?

Target word: p-value and its meaning **Decides:** 3 of 1,000 simulated results were at least as extreme.

A	Correct	A result this extreme almost never happens with a fair coin.
B	Absolute language	A p-value measures evidence; it never proves.
C	Wrong stem number	34/50 is the sample proportion of heads, not a p-value.
D	Inverted operation	Counts the less extreme results instead.

9. Answer C Concept 6 · transfer

In 500 simulated trials that assume a claim is true, a result at least as extreme as the observed one happened 60 times. What is the approximate p-value?

Target word: p-value from simulation counts **Decides:** 60 / 500.

A	Confusion set	$500 \div 60$ inverts the fraction; probabilities cannot exceed 1.
B	Inverted operation	The share of results less extreme.
C	Correct	$60 \div 500 = 0.12$.
D	Dropped element	A count, not a proportion.

10. Answer B Concept 7

~~In a class of 30, a survey taken in October~~ shows $P(\text{plays soccer}) = 0.4$, $P(\text{in band}) = 0.3$, and $P(\text{both}) = 0.1$. What is $P(\text{soccer or band})$?

Target word: $P(A \text{ or } B)$ **Decides:** Subtract the overlap once.

A	Confusion set	Multiplies, which is the “and” rule for independent events.
B	Correct	$0.4 + 0.3 - 0.1 = 0.6$.
C	Dropped element	Leaves out the overlap, counting “both” twice.
D	Inverted operation	Adds the overlap instead of subtracting it.

11. Answer B Concept 7 · transfer

$P(A \text{ or } B) = 0.75$, $P(A) = 0.5$, and $P(B) = 0.45$. What is $P(A \text{ and } B)$?

Target word: solve for the overlap **Decides:** $0.75 = 0.5 + 0.45 - P(A \text{ and } B)$.

A	Wrong stem number	$0.75 - 0.45$ subtracts only one probability.
B	Correct	$0.95 - 0.75 = 0.20$.
C	Adjacent category	0.5×0.45 assumes independence, which is not given.
D	Dropped element	The sum of $P(A)$ and $P(B)$, before removing $P(A \text{ or } B)$.

12. Answer D Concept 8

Using foam dice ~~from the supply closet~~, a student says a sum of 7 and a sum of 12 are equally likely, since each is just one possible sum. ~~The class has a double period that day.~~ What is the teacher’s best response?

Target word: addressing equiprobability bias **Decides:** The sample space shows 6 ways to make 7 and 1 way to make 12.

A	Adjacent category	Gives the answer without confronting the student’s reasoning.
B	Confusion set	Equally likely faces do not make equally likely sums.
C	Dropped element	Ten rolls may not show the pattern; it needs many trials or the sample space.
D	Correct	The sample space makes the 6-to-1 difference visible.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Skew pulls the mean	In right-skewed data the mean is pulled above the median toward the long tail; the median better represents a typical value.	Reporting the mean for skewed data because it “uses every value.”	A few very large (or very small) values in otherwise clustered data.
2	Linear transformations of data	Adding a constant shifts the mean but leaves the standard deviation unchanged; multiplying by k multiplies both the mean and the standard deviation by $ k $.	Thinking adding points increases spread, or scaling the SD by k^2 .	“Every value is increased by / multiplied by...”
3	The $1.5 \times \text{IQR}$ outlier rule	$\text{IQR} = Q3 - Q1$; values below $Q1 - 1.5 \cdot \text{IQR}$ or above $Q3 + 1.5 \cdot \text{IQR}$ are outliers.	Using $1 \times \text{IQR}$, or calling anything beyond $Q1$ or $Q3$ an outlier.	Quartiles given and “which value is an outlier?”
4	Observational studies cannot show cause	Only a randomized experiment supports a cause-and-effect conclusion; observational data show association that may come from confounding variables.	Accepting causation because the sample is large.	A survey or records study followed by a claim that X “causes” or “improves” Y .
5	Categorical vs. quantitative variables	A variable is quantitative only if arithmetic on it makes sense; numbers used as labels (ZIP codes, jersey numbers) are categorical.	Calling any variable recorded as a number quantitative.	A list of variables that includes numeric labels.
6	Simulation-based p-values	Simulate many samples assuming the claim is true; the p-value is the proportion of simulated results at least as extreme as the observed one.	Using the observed sample proportion as the p-value, or treating a small p-value as proof.	“Simulated 1,000 times assuming...”
7	The addition rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.	Adding without subtracting the overlap, or multiplying.	“Or” with events that can happen together.
8	The equiprobability misconception	Outcomes are equally likely only when each can happen in the same number of ways; listing the sample space shows why some sums are more likely.	Believing every possible result of a random process has the same chance.	A student says two outcomes are “equally likely” because each is possible.

Course scope (public description): Statistics for Secondary Mathematics Teaching provides a comprehensive exploration of advanced concepts and methodologies in teaching statistics and probability. The course delves into key topics such as summarizing and representing data, study design, sampling, testing claims, drawing conclusions, and probability. Emphasizing a deep understanding of conceptual foundations, including their historical development and perspectives from diverse cultures, the course addresses common misconceptions and students' ways of thinking. The appropriate use of instructional practices is also highlighted. There are no prerequisites for this course.